

Math 125 Midterm Review 1 Solutions

Find the following limits (without using L'Hopital's Rule)

1. Fall 2015 Q1)

(a)

$$\lim_{x \rightarrow 3} (2x^2 - 5x - 3)/(x^2 + 4x - 21) = 7/10$$

(b)

$$\lim_{x \rightarrow 0} \sin^2(5x)/(x \sin(2x)) = 25/2$$

(c)

$$\lim_{x \rightarrow \infty} \sqrt{x^4 + x^2 + 5} - x^2 = 1/2$$

2. Fall 2011 Q1)

(a)

$$\lim_{x \rightarrow 0} (\sin x + \cos(1/x))x^2 = 0$$

(b)

$$\lim_{x \rightarrow 1} (x^2 - 1)/|x - 1| \text{DNE}$$

(c)

$$\lim_{x \rightarrow -\infty} x + \sqrt{x^2 + 2x + 3} = -1$$

3. Fall 2018 Q1 (parts a-c)

(a)

$$\lim (2x^2 - 3x - 2)/(x - 2) = 5$$

(b)

$$\lim x/(x + \sin 2x) = 1/3$$

(c)

$$\lim \sqrt{x^2 + 4x} - x = 2$$